MILITARY SPECIFICATION SHEET

ELECTRON TUBE, STORAGE

TYPE 7571 1/

This specification sheet forms a part of the latest issue of Military Specification MIL-E-1.

DESCRIPTION: Recording, magnetic deflection and magnetic/electrostatic focus (See note 1)

CATHODE: Coated unipotential --- Outline and pin connections (See figure 1)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb1	Eb2	EL1	EL2
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc
Maximum:	6.6	0, -100	700	4,500	4,500	1,000	1,000
Minimum:	6.0	---	---	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	400	3,500	Adjust	750	Adjust

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	E Decel	E Storage screen	Ehk	Rg1	E Collector
Unit:	Vdc	Vdc	Vdc	Ohms	Vdc
Maximum:	1,000	1,000	+10	500 K	1,000
Minimum:	---	---	-125	0	0
TEST CONDITIONS:	750	Adjust	---	---	750

REQUIREMENT OR TEST:

GENERAL

3.2	Qualification - Required	E-50.3.1	Preheating (t = 180 sec.)
E-50.2	Holding period	Section 5	Preparation for delivery - package group 4, container size (L), (i)

1/ Formerly CK7571/QK685

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification							
3.5.2	Plastic Base material	See note 3	---	---	---	---	---	---
	Quality conformance inspection, part 1							
5201	Voltage breakdown		---	---	---	---	---	---
5201	Voltage breakdown (Magnetic types)		---	---	---	---	---	---
5251	Heater-cathode leakage		---	---	---	---	---	---
5251	Grid No. 1 leakage		---	---	---	---	---	---
5251	Grid No. 2 leakage		---	---	---	---	---	---
5251	Anode No. 1 leakage		---	---	---	---	---	---
5251	Anode No. 2 leakage		---	---	---	---	---	---
---	Grid cutoff voltage	See notes 1 and 2	---	---	Eco	-15	-50	V
1256	Cathode current	Ec1 = 0 (See note 5)	---	---	Ik	300	---	uA
---	Electrode current (Beam)	Ec1 = 0 (See note 6)	---	---	Ib	8	---	uA
---	Modulation	Ib = 2 uA (See note 7)	---	---	ΔEc1	---	10	V
---	Signal to shading ratio	90% st diameter (See notes 8 and 9) EL2 = 225 to 425 V	---	---	ratio	5:1	---	---
---	Overall signal to shading ratio	100% st diameter (See notes 8 and 9) EL2 = 225 to 425 V	---	---	ratio	3:1	---	---
---	Blemishes	See notes 8 and 10	---	---	---	---	---	---
---	Resolution (Magnetic)	Read Ib = 1uA (See notes 8 and 12) Eb2 = Eb1 = 3,500 V	---	---	---	1000	---	Lines
---	Resolution (Electrostatic)	(See notes 8 and 12) Eb2 = 120 V to 420 V (Vary for optimum focus) Eb1 = 3,500 V	---	---	---	700	---	Lines
---	Dynamic range (1)	See notes 8 and 13	---	---	---	---	5	Steps
---	Dynamic range (2)	See notes 8 and 14	---	---	---	---	---	---
---	Writing speed	See notes 8 and 15	---	---	---	2	---	Fields
---	Integration	See notes 8 and 16	---	---	---	50	---	Fields
---	Priming speed	See notes 8 and 17	---	---	---	10	---	Fields
---	Spurious emission	See notes 8 and 18	---	---	---	---	---	---
---	Static storage	See notes 8 and 19	---	---	---	---	---	---
---	Ion erasure	See notes 8 and 20	---	---	t	36	---	---
---	Storage dielectric critical voltage (First cross over)	See notes 8 and 20 (a)	---	---	---	40	---	V

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (Cont)							
---	Storage dielectric breakdown voltage	See notes 8 and 20 (a)	---	---	---	40	---	V
	Quality conformance inspection, part 2							
1331	Direct-interelectrode capacitance	Collector electrode to all others	6.5	L6	C	10	20	pf
		Storage screen to all others	6.5	L6	C	25	45	pf
		Grid No. 1 to all others	6.5	L6	C	10	20	pf
1301	Heater or filament current		6.5	L6	If	540	660	mA
---	Resolution (2) Magnetic	Read Ib = 1 uA (See notes 8 and 11)	---	---	---	1000	---	Lines
1105	Permanence of marking		---	---	---	---	---	---
Appendix D, 20 (b)	Dimensions	See figure 1	---	---	---	---	---	---
	Quality conformance inspection, part 3							
4.7	Life-test provisions	Group D; t = 500 hr minimum (See note 21)	---	---	---	---	---	---
4.7.3.1	Cathode-ray tubes, end points	See note 22	---	---	---	800	---	Lines
1136	Container drop (Cathode ray tubes) (Periodic check)	Rough handling test (i) Test c, d, and e (See note 23)	---	---	---	---	---	---

NOTES:

1. Deflection drive, both horizontal and vertical, must be applied whenever the tube is conducting to avoid damage to the storage surface. With 3,545 volts dc on the storage screen, decelerator, lenses and collector connected in parallel, and with all other tube voltages at their specified test levels, the current measured from the storage assembly will be the total primary beam current.

2. Grid cutoff voltage is a grid No. 1 potential such that the corresponding beam current is less than or equal to .05 microampere and greater than 0 microampere.

3. With the qualification approval test report, the tube manufacturer shall certify that the base used has been fabricated from an insulating material qualified under MIL-M-14 for type CFG (phenolic) or MFE (low-loss phenolic). Bases previously qualified to method 1216 of MIL-E-1 are acceptable.

4. At the completion of the container drop test the resolution of the tube shall be checked. The resolution shall be a *minimum of 1,000 lines as measured with the procedure set forth in notes 8 and 11.*

5. With all other electron gun potentials at their specified test levels and with the grid No. 1 potential at 0 volts, the cathode current shall be within the limits specified. During this test, both horizontal and vertical scan must be on and the beam must be defocussed. The gun must not be left in this 0 bias condition for more than 30 seconds.

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NOTES: (Cont)

6. With the storage assembly connected as specified in note 1, with all other electron gun potentials at their specified test levels, and with the grid No. 1 potential at 0 volts, the total primary beam current measured from the storage assembly will be within the limits specified. During this test, both horizontal and vertical scan must be on and the beam must be defocussed. The gun must not be left in this 0 bias condition for more than 30 seconds.

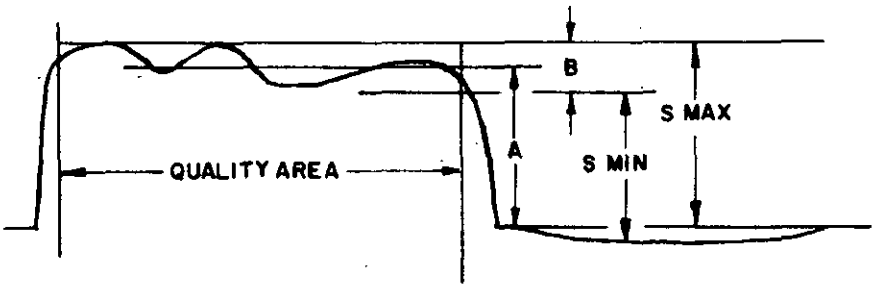
7. With the storage assembly connected as specified in note 1 and with the grid No. 1 potential at grid cutoff voltage, the change in grid No. 1 potential to produce a beam current of 2 microamperes shall be within the limits specified.

8. The storage tube output for this test shall be monitored visually on a television display as well as electrically on an oscilloscope displaying the read out of lines in the output scan, hereafter known as an "A" scope. A 525 line interlaced TV raster conforming to United States FCC television standards and a Model 545 Tektronix Oscilloscope, or equivalent, shall be used. With the storage tube considered as the first stage of the video system, the read out video amplifier should have a bandwidth such that its response is flat to 1 db out to 6.5 Mc to insure accurate display of the resolution capabilities of the tube. For the dynamic tests to follow, the storage tube shall be operated in four modes: (1) erase, (2) prime, (3) write, and (4) read. The focus electrode shall be operated at anode potential unless specified otherwise. The decelerator, lens No. 1, and collector voltages shall be at 750 volts dc, lens No. 2 will be at 300 volts dc initially. The optimum voltage for this lens will be determined during the test for "Signal to Shading Ratio". The grid No. 1 bias shall be set for each mode initially so as to provide the following cathode currents: (1) erase - 50 uA, (2) prime - 50 uA, (3) write - beam current cutoff, and (4) read - 5 uA. The storage screen voltage settings shall be as follows: (1) erase - 90 to 100 percent decelerator voltage, (2) prime - 20 volts dc, (3) write - 50 to 90 percent decelerator voltage, and (4) read - adjust for proper black background level (usually between 12 to 16 volts dc). The input video signal for write should be clamped to grid cutoff and driven in a positive direction. The amplitude of the write input video signal must be determined empirically during tests and will be dependent upon writing time and the transfer characteristic of the electron gun of the tube being tested.

9. The signal-to-shading ratio shall be defined as the ratio of the average signal amplitude to the peak-shading variation of the signal. (See figure diagram below.)

   The tube shall be set up with the television raster expanded in both horizontal and vertical directions such that the complete area of the storage surface is inscribed within a square raster. Shading and background uniformity shall be optimized during this test by the proper setting of collimation lens voltage settings, focus coil position and the currents in beam centering coils which may be used.

   Signal to shading tests will be accomplished in the prime mode of operation. The storage screen voltage during prime will be adjusted for 20 volts dc. Initially, grid No. 1 bias settings will be adjusted to provide 50 to 100 microamperes of cathode current and the beam will be defocussed to provide uniform coverage of the storage area by the raster lines until the tube is fully primed. As soon as a full prime has been accomplished, the grid No. 1 bias shall be reset for a cathode current of 5 microamperes and the beam shall be refocussed. When the variables stated above (collimation lens voltage, focus coil position, etc.) have been adjusted for optimum output uniformity, the signal to shading ratio defined above will be measured on the "A" scope presentation.



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NOTES: (Cont)

9. (Cont)

Figure diagram (Cont)

$$A = \frac{Smax. + Smin.}{2}$$

$$B = Smax. - Smin.$$

$$\text{Signal/Shading Ratio} = \frac{A}{B}$$

(Blemishes and ion spot shall be ignored in this measurement).

The quality area is defined as the storage screen area enclosed in a circle centered on the storage screen and having a diameter equal to 90 percent of the storage screen diameter.

10. With the raster scan expanded such that the storage area is inscribed, the storage screen will be visible on the TV monitor as a circular area and blemishes will be seen as bright and dark spots. The following table describes the acceptance limits of all blemishes. Blemish size shall be defined in terms of the average diameter of the blemish as seen on the monitor as a percentage of the diameter of the storage area as seen on the monitor. Blemish amplitude shall be determined by examining on an oscilloscope the video signal from a horizontal scan through the blemish. Blemish amplitude shall be expressed as a percentage of the reference white output signal obtained when reading at the same storage screen voltage used for prime. The screen shall be erased and primed before inspecting for blemishes.

Inspection for white blemishes shall be made with the storage screen voltage set just at cutoff for the area being examined. Black blemishes shall be counted with the storage screen voltage halfway between the prime voltage and read storage screen cutoff voltage.

Class	Size %	Polarity	Minimum Amplitude %	Maximum No.	
				Within quality area	Anywhere
A	.125-.250	white	20	15	25
B	.250-.50	white or black	20	7	10
C	.50-1.0	white or black	20	2	4
D	1.0-100	white or black	20	0	1
Total A+B+C+D				15	25

11. This resolution test shall be made using a series of equally spaced horizontal lines written with an unmodulated beam as the write pattern. The lens voltages, focus coil position, and beam centering coil currents shall be the same as used during measurement of signal-to-shading ratio. The scan during read shall be at right angles to the written pattern with the output signal monitored on an oscilloscope. A reference signal amplitude shall be determined by writing widely spaced lines. The line spacing is then reduced until the read peak-to-peak signal is 50 percent of the reference amplitude. The TV lines resolution shall be computed as follows:

$$\text{Number of lines written} \times 2 \times \frac{\text{Diam. of stor. screen}}{\text{Height of written line pattern}} = \text{TV lines resolution}$$

Without readjusting focus, the tube should be capable of meeting this specification with the write scan rotated 90°.

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NOTES: (Cont)

12. Resolution will be measured by use of a resolution test pattern such as obtained from a 2F21 monoscope, or equivalent, signal source. The output signal from the storage tube will be observed on a TV monitor. The lens voltage, focus coil position, and beam centering coil currents shall be the same as used for signal-to-shading measurements. The write scan size shall be adjusted so that the diameter of the large circle in the test pattern is equal to one-third the diameter of the storage screen. The limiting resolution determined from the TV monitor shall be multiplied by three to obtain resolution for the full diameter. Without readjusting focus, the tube should be capable of meeting this specification with the write scan rotated 90°.

13. The tube shall be capable of storing and reading out onto a monitor the test pattern from the 2F21 monoscope, or equivalent, signal source showing a gray scale wedge with a minimum of five distinguishable steps.

14. The difference in storage screen voltage between prime and read cutoff, at center of the tube, shall be at least 15 percent of the prime voltage. Ample time shall be allowed for priming.

15. The tube shall be set up so that the raster height and width are just equal to the diameter of the storage screen surface. The standard TV test pattern shall be written and stored to a 100 percent modulation level in the specified number of fields. The raster shall be the standard 525 line system.

16. With the test pattern and raster size used in note 15, the pattern shall be written in 50 fields of a television raster. Input video level shall be reduced to such a level as to prevent over-writing of white areas on the picture.

17. The test of priming speed shall be done with the beam defocussed and the deflection size as in note 15. The screen should be erased then primed for the specified number of TV scans using a cathode current that corresponds to a beam current of 5 microamperes. The storage screen cutoff voltage shall then be at least 90 percent of the cutoff voltage measured when ample time is allowed for prime.

18. The tube should be erased, primed at 20 volts, and then switched to the write mode with the control grid (G1) voltage below cutoff. The deflection yoke should then be disconnected. After 30 seconds, the yoke should be reconnected and the storage screen voltage reduced to zero. The control grid bias should then be adjusted to produce a cathode current of 5 microamperes. Raise storage screen voltage until entire screen just becomes visible on the TV monitor. Note any buildup of signal on the background level. This will be the result of spurious emission. The amplitude of the buildup signal shall be no more than 10 percent of the output obtained with the storage screen at 20 V.

19. Static storage should be tested in the following fashion. The storage screen shall be uniformly primed to a voltage of 20 volts and then the storage screen read cutoff determined and recorded. The gun shall be biased off and then all voltages turned off. At the end of 5 minutes when the tube tube is turned on, the voltage required for storage screen cutoff shall not have shifted by more than 2 volts.

20. Before conducting this test, the operating conditions for 0.1 microampere collector output current during read must be determined. The scan should be reduced so that only the center area (approximately 70 percent) of the storage screen is scanned. The tube should be erased then switched directly to read with the storage screen at 16 volts. After the screen has primed to this voltage, the control grid bias should be adjusted to produce 0.1 microampere current.

The tube should then be erased, primed at 20 volts, and switched to read with the storage screen voltage set just at cutoff. A maximum ion spot build up amplitude equal to 25 percent of average signal amplitude (see note 9) is permitted 36 seconds after switching from prime to read mode while reading at the control grid bias established above.

20(a). The tube shall be erased, primed at least 40 volts, and switched to read with the screen voltage set just at cutoff. Then the tube should be erased, primed at 40 volts, and written with the test pattern. The pattern should be readable without evidence of breakdown.

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NOTES: (Cont)

21. Operational modes shall be automatically switched from ptime to erase at a rate not to exceed one cycle per minute. The tube shall be in each mode 40 percent to 60 percent of the time. Television raster scan shall be used such that the storage area is inscribed within the raster. The cathode current for erase and prime shall not exceed 50 microamperes.

22. Tests made to determine life-test end point shall be as specified in 4.7.3.1 and magnetic focus resolution as in notes 8 and 11, but with a minimum value as specified for this test.

23. This test is not required for qualification approval.

Custodians:

Army - EL
Navy - SH
Air Force - 11

Review activities:

Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:

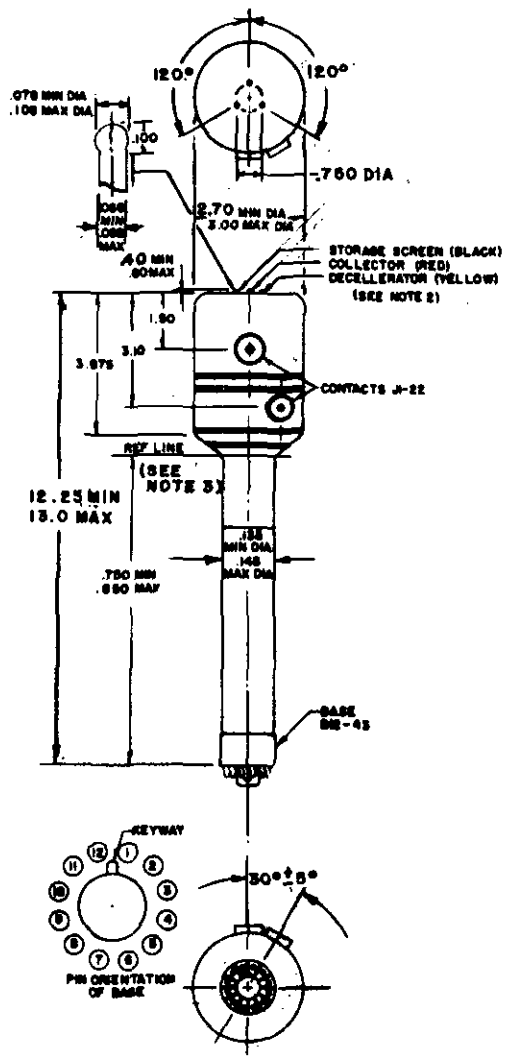
Army - MU
Navy - WP, MC, CG
Air Force - None

Preparing activity:

Navy - SH

(Project 5960-1926)

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Pin no.	Element
1	Heater
2	Grid 1
3	No connection
4	Focus electrode (A2)
5	No connection
6	Anode (A1)
7	No connection
8	No connection
9	No connection
10	Grid 2
11	Cathode
12	Heater

INCHES	MM	INCHES	MM
.068	1.73	.75	19.05
.078	1.85	1.50	38.10
.088	2.24	2.70	68.58
.100	2.54	3.00	76.20
.108	2.74	3.100	78.74
.135	3.44	3.875	98.43
.145	3.68	7.50	190.50
.400	10.17	8.50	215.90
.550	13.97	12.25	311.15
.600	15.24	13.00	331.20

NOTES:

1. The socket for this base is not to be rigidly mounted. It must have flexible leads and must move freely. The bottom circumference of the base will fall within a circle concentric with cone axis and having a diameter of 2.00 (50.8) inches.
2. The top pin connections should have flexible leads and be allowed to move freely.
3. Reference line is determined by the position where a reference line gage .500+.003(12.70+.08)I.D. and 2.00(50.8) long will rest on the bulb cone.
4. Nominal dimensions are for information purposes only and are not required for inspection purposes.
5. Metric equivalents are for general information only and are based upon 1 inch = 25.4 mm to the nearest .01 millimeter.
6. For quality conformance inspection, Part 2, dimensions are to be inspected with an AQL of 6.5 percent and an inspection level of L6.

Figure 1. Outline drawing of electron tube type 7571.

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